Science

Nature Notebook
Nature Walks & Scouting
Food Science Lessons
Food Science Labs

Alveary

Alt. Science Options

Science: Food Science

GRADE(S): 9-12

About the Course

This course includes the following topic(s): Food Science Lessons, Food Science Labs, Nature Notebook: Grades 9-12, Nature Walks & Scouting: Grades 9-12

About Nature Notebook: Grades 9-12
Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

About Nature Walks & Scouting: Grades 9-12
Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

About Food Science Lessons
An elective in Physical Science, Food Science introduces learners to applications in chemistry and physics as they relate to cooking and baking, while incorporating historical context, modern developments, current events, and citizenship. This is a hands-on course that requires independent interest and some flexibility for work time in the kitchen. There is room for teachers and learners to adjust the course for personal needs and preferences.

About Food Science Labs
Note that labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

Placement & Combining Tips

Nature Notebook: Grades 9-12
Learners may be combined and follow their own interests.

Nature Walks & Scouting: Grades 9-12
Learners may follow their own interests or follow the plan of their local scouting troop or natural history club.

Food Science Lessons
There are no specific prerequisites or concurrents for this course.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12	Nature Notebook: Grades 9-12 1+ time/week 20 min+	
9-12	Nature Walks & Scouting: Grades 9-12 1 time/week 30 min+	
9-12	Food Science Lessons 5 times/week 45 min	Salt, Fat, Acid, Heat: Mastering the Elements of Good Cooking The Elements of Baking Gullah Geechee Home Cooking: Recipes from the Matriarch of Edisto Island Mexico in Your Kitchen Classic Russian Cooking Pomp and Sustenance Celtic Folklore Cooking Culinary Reactions: The Everyday Chemistry of Cooking

GRADE	SCHEDULE INFO.	BOOKS
9-12	Food Science Labs 1 time/week 60 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Food Science				
Food Science Lessons Nature Walks & Scouting: Grades 9-12	Food Science Lessons	Food Science Lessons	Food Science Lessons Nature Notebook: Grades 9-12	Food Science Lessons Food Science Labs

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Food Science Lessons
☐ Carve out time to continue or establish the regular habit of spending time in nature, including the use of a nature journal, as appropriate.

☐ Obtain materials from the supply lists.

☐ The lab period is the student's opportunity to prepare a meal (or part of a meal, depending on teacher preference). Therefore, it should be scheduled at a time when there will be people to eat what they prepare, which may not be during their school day.

☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.

☐ Bookmark or print Quick Links, as needed.

Term Prep & Teacher Tips

Food Science Lessons
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
• basic cookware and bakeware (pots, pans, baking sheets, mixing bowls)
• basic cooking utensils (mixing spoons, spatula, tongs, knives, measuring cups and spoons, whisk, meat thermometer)
• kitchen scale
• mortar and pestle
• salt and pepper
• romaine lettuce (Term 1 Week 1)
• anchovies (Term 1 Week 1)
• mayonnaise or an egg to make fresh (Term 1 Week 1)
• garlic clove (Term 1 Week 1)
• lemon juice (Term 1 Week 1)
• white wine vinegar (Term 1 Week 1)
• parmesan cheese (Term 1 Week 1)
• Worcestershire sauce (Term 1 Week 1)
• olive oil (Term 1 Week 1)
• bread of your choice (Term 1 Week 1)
• dinner protein of your choice (Term 1 Week 1)
• ingredients by student choice (students are prompted to make a grocery list most weeks)

• optional: additional kitchen tools as indicated in cookbooks, depending on recipes chosen by students

Reminders

Food Science Lessons
☐ Note that Lab in week 1 calls for lettuce, anchovies, mayonnaise/egg, garlic clove, lemon juice. white wine vinegar, parmesan cheese, Worcestershire sauce, olive oil, and bread and dinner protein of choice. Additional grocery lists will be provided by the student throughout the year.

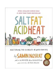
Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details]

Science: Food Science

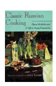
Food Science Lessons

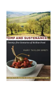

Salt, Fat, Acid, Heat: Mastering the Elements of Good Cooking


The Elements of Baking


Gullah Geechee Home Cooking: Recipes from the Matriarch of Edisto Island


Mexico in Your Kitchen


Classic Russian Cooking


Pomp and Sustenance


Celtic Folklore Cooking


Culinary Reactions: The Everyday Chemistry of Cooking

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies]

(No Subject Supplies Assigned)

Quick Links

Click THIS text or scan the QR code for links.

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Food Science Lessons

∞ [America's Test Kitchen Library]

	∞ The Loopy Whisk ∞ Samin Nosrat Cooking ∞ Food & Wine ∞ Science News Explores

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How To Approach

Introduce

• If starting a new book or a new topic in the book, then look at the title or a picture and take a moment to consider previous ideas and experiences.

• If continuing a previous reading, recap what was read previously. Often, the title of the book's section can help to draw out the main idea.

Read

• Read or do, as instructed in the lessons, making note of or flagging unfamiliar terms, interesting ideas, important dates, inspiring quotes, etc.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.

• If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab/field work further supports major concepts from the text.

Narrate

• Process the ideas of the lesson by retelling, defining a concept, explaining the links in a chain of thought, etc. Do this orally or silently to yourself.

• Use words, pictures, outlines, etc.

• If a particular idea cannot be narrated, then read or examine the text again.

Discuss

• Consider with the teacher any thoughts, confusion, or concerns about the passage.

• If understanding is still uncertain, try rereading the passage or do some personal research on the topic.

Connect

• Follow any extra links, examine any sidebars in the text, or pursue additional reading, depending on student interest.

Science: Food Science

How To Approach

Introduce

• Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.

• Relevant concept(s) are introduced in the text.

• Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.

Lab Procedure

• Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.

• Record all data and observations in the lab notebook.

Analysis & Conclusions

• After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.

• If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

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Term 1

WEEK 1

⬚ **45m Food Science Lessons - Lesson 1**
Preparation

Materials: Video Link

→ INTRO
Your lessons are organized to teach you the basics in Term 1 and then encourage exploration in Terms 2-3. If you prefer to spread out the readings in Term 1, there is room to do so. Feel free to work at your own pace.

As we start our kitchen adventure, it is important to remember that we are also investigating as scientists. Let's watch a portion of this public lecture from the Science & Cooking lecture series at Harvard to get us in the mindset that the best chefs must be scientists, too.

→ VIEW, NARRATE, & DISCUSS
Watch a portion of the first Science and Cooking lecture:
∞ Video Link: Science and Cooking (5:45-57:14)

• PLAN WEEKLY
⬚ take a nature walk
⬚ science free read

WEEK 1

⬚ **45m Food Science Lessons - Lesson 2**
The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ INTRO
In this book, a chef leads students through 4 important "elements" of cooking: salt, fat, acid, and heat. The first 3 are families of chemical substances, while the last is really about adding energy. If you have already taken Chemistry, think about what you've already learned about salt, fat, acid, and heat as you observe the roles they play in cooking. If you haven't yet taken Chemistry, you'll learn a few basics here and will get to make lots of observations that will support any future study in the subject.

→ READ, NARRATE, & DISCUSS
p.17-28

∞ Video Link: America's Test Kitchen
∞ Video Link: Alveary Chemistry Companion

WEEK 1

⬚ **45m Food Science Lessons - Lesson 3**
The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS
p.29-39

WEEK 1

⬚ **45m Food Science Lessons - Lesson 4**
The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS
p.40-55

WEEK 1

⬚ **45m Food Science Lessons - Lesson 5**
The Science of Cooking & Baking

Materials: Salt, Fat, Acid, Heat, Elements of Baking

→ PREPARE
Choose a ‘Salt Lesson’ recipe from p.428 of SFAH for next week. Then decide what else you will serve to balance the meal. Healthy Eating Plate (linked) may help; for more information about the history of dietary guidelines, the newly released U.S. guidelines, or Canadian guidelines, check out the other optional links. Be sure to make a shopping list for your teacher.
∞ Website Link: Healthy Eating Plate
∞ Website Link: Illustrating Diet Advice is Hard
∞ Website Link: Dietary Guidelines for Americans
∞ Website Link: Canada's Dietary Guidelines

Then begin Elements of Baking:

→ INTRO
In the second text for this course, a chemist leads students through studies in baking with the mindset of a materials chemist: we observe the physical properties of various materials and experiment with how they perform in various situations to work toward a desired product. The approach is the same whether we are developing new adhesives for space shuttle tiles, new polymers for biomedical applications or textiles, or new recipes for friends and family.

Note the organization of EB on p.13. Feel free to flip around so that you understand how the book is organized and where to find what you want.

Read p.15. Note that you will measure by weight in this course. In science, we experiment, and the accuracy and precision of weight measurements will ensure that we can draw conclusions from our experiments.

→ READ, NARRATE, & DISCUSS
p.19-23

WEEK 1

⬚ **60m Food Science Labs - Lesson 1**

> Materials: Salt, Fat, Acid, Heat, ingredients for Caesar Salad p.48, protein, bread, indicated kitchen supplies

→ KITCHEN LAB
We'll learn more about salads in the coming weeks, but for this week, make Caesar Salad as on p.48-49 in SFAH. You may either use store-bought mayo for now or try the recipe (visual p.86-87 or traditional p.375). Serve alongside your choice of protein and bread/croutons. If you want to make your own mayonnaise, you might find the video below helpful.
∞ Video Link: Mayonnaise and the Science of Emulsions

WEEK 2

⬚ **45m Food Science Lessons - Lesson 6**
The Science of Baking

> Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS
Ch.2 p.37-44

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 2

⬚ **45m Food Science Lessons - Lesson 7**
The Science of Baking

> Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS
Ch.3 p.48-49. Read once before case studies, but you will come back to them after (or even during) case studies as you develop understanding. Make sure you understand WHY the points on p.49 are so.

→ PLAN
Look over p.50-51: In addition to Buttery Vanilla Cake, select 3 items from the pink boxes which you are familiar with or interested in (except bread, which we'll get to later, and waffles).

→ PREPARE
Read over the rules for Buttery Vanilla Cake (p52-54) and your 3 additional items, noting how the rules apply the science of ingredients from Ch.2, how the vegan and gf-vegan rules combine the first 3, and how the rules are the same/different between items.

Refer to p.91 for a Table of Contents of case studies to locate the recipe pages for your 3 selections. Note that a table of the ingredients needed for different variations is printed toward the end of the case study (p.104 for Buttery Vanilla Cake). You can choose whatever variations most interest you, but the key to learning the science is to try different variations because we have to change a variable to see what happens. Add any needed ingredients to your shopping list.

WEEK 2

⬚ **45m Food Science Lessons - Lesson 8**
The Science of Baking

> Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS
Ch.4 p.88-89.
Buttery Vanilla Cake p.92-105. Even if you aren't planning to do all of the variations, reading about the role the ingredients play is important!

WEEK 2

⬚ **45m Food Science Lessons - Lesson 9**
The Science of Baking

> Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare Buttery Vanilla Cake, recording what you do and what you observe in your notebook.

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Term 1

WEEK 2

⬚ 45m Food Science Lessons - Lesson 10

The Science of Baking

Materials: Elements of Baking

→ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week.
• Add frostings and creams from Ch.10, if desired.
• Add to shopping list, as needed.
• If you want to decide on your Salt Lesson recipe for next week, you could do this now, or you could wait until after dinner night. You could try one of these first two again or try a new one. Add to shopping list, as needed.

WEEK 2

⬚ 60m Food Science Labs - Lesson 2

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB
Prepare the chosen Salt Lesson. Decide on next Salt Lesson p.428 and add to shopping list, if you haven't already.

WEEK 3

⬚ 45m Food Science Lessons - Lesson 11

The Science of Baking

Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS
Ch.5 p.259-266, p.267-268 optional.

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 3

⬚ 45m Food Science Lessons - Lesson 12

The Science of Baking

Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS
Ch.5 p.269-275

WEEK 3

⬚ 45m Food Science Lessons - Lesson 13

The Science of Baking

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare Buttery Vanilla Cake, including any changes or adjustments noted last week. Record what you do and what you observe in your notebook.

WEEK 3

⬚ 45m Food Science Lessons - Lesson 14

The Science of Baking

Materials: Elements of Baking

→ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next case study by reading any introduction, as well as the recipe (you will have more time to finish reading later, but don't wait to add to the shopping list, as needed). Even though you have read about bread in the gluten-free chapter, complete your case studies before moving on to bread, as this brings in new material to consider (psyllium husk).

WEEK 3

⬚ 45m Food Science Lessons - Lesson 15

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS
p.205, 208-223

→ PLAN
Consider your Salt Lesson for this week. Will you use any of the ideas in this reading for the recipe? If not, you can always come back to them in a future recipe.

If you want to decide on your Salt Lesson recipe for next week, you could do this now, or you could wait until after dinner night. You could try one of these first two again or try a new one. Keep in mind that any salad can be a Salt Lesson, so feel free to experiment with the Ideal Salad (p.216) or Avocado Matrix (p.222). Add to shopping list as needed.

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Term 1

WEEK 3

⬚ **60m Food Science Labs - Lesson 3**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Salt Lesson. Decide on next Salt Lesson p.428 and add to shopping list, if you haven't already.

WEEK 4

⬚ **45m Food Science Lessons - Lesson 16**
The Science of Baking

Materials: Elements of Baking

➔ READ, NARRATE, & DISCUSS
Finish any reading from last week, then read Ch.6 p.311-317.

➔ PREPARE
Complete any preparation that you need to do for your bake tomorrow.

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 4

⬚ **45m Food Science Lessons - Lesson 17**

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

➔ BAKING DAY!
Prepare Buttery Vanilla Cake, including any changes or adjustments noted last week. Record what you do and what you observe in your notebook.

WEEK 4

⬚ **45m Food Science Lessons - Lesson 18**
The Science of Baking

Materials: Elements of Baking

➔ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.

WEEK 4

⬚ **45m Food Science Lessons - Lesson 19**
The Science of Cooking

Materials: Salt, Fat, Acid, Heat

➔ READ, NARRATE, & DISCUSS
p.57-71

∞ Video Link: America's Test Kitchen
∞ Video Link: Alveary Chemistry Companion

WEEK 4

⬚ **45m Food Science Lessons - Lesson 20**
The Science of Cooking

Materials: Salt, Fat, Acid, Heat

➔ READ, NARRATE, & DISCUSS
p.72-87

➔ PLAN
Consider your Salt Lesson for this week. Will you use any of the ideas in this reading for the recipe? If not, you can always come back to them in a future recipe.

If you want to decide on your Cooking Lesson recipe for next week, you could do another Salt Lesson or try a Fat Lesson (p.429). You could wait to decide until after dinner night, too. Add to shopping list, as needed.

WEEK 4

⬚ **60m Food Science Labs - Lesson 4**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Salt Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

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Term 1

WEEK 5

⬚ **45m Food Science Lessons - Lesson 21**
The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ **READ, NARRATE, & DISCUSS**
p.88-99

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 5

⬚ **45m Food Science Lessons - Lesson 22**
The Science of Baking

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ **BAKING DAY!**
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 5

⬚ **45m Food Science Lessons - Lesson 23**
The Science of Baking

Materials: Elements of Baking

→ **ANALYZE**
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.

WEEK 5

⬚ **45m Food Science Lessons - Lesson 24**
The Science of Baking

Materials: Elements of Baking

→ **READ, NARRATE, & DISCUSS**
Ch.7 p.343-349

WEEK 5

⬚ **45m Food Science Lessons - Lesson 25**
The Science of Baking

Materials: Elements of Baking

→ **READ, NARRATE, & DISCUSS**
Ch.7 p.350-355. Compare the application here, but don’t worry about mastering or remembering it. You can always come back as you try particular recipes.

→ **PLAN**
Decide on your Cooking Lesson recipe for next week, or wait to decide until after dinner night. Add to shopping list, as needed.

WEEK 5

⬚ **60m Food Science Labs - Lesson 5**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ **KITCHEN LAB**
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven’t already.

WEEK 6

⬚ **45m Food Science Lessons - Lesson 26**
The Science of Baking

Materials: Elements of Baking

→ **READ, NARRATE, & DISCUSS**
Ch.7 p.356-360

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 6

⬚ **45m Food Science Lessons - Lesson 27**
The Science of Baking

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ **BAKING DAY!**
Prepare your selected recipe. Record what you do and what you observe in your notebook.

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Term 1

WEEK 6	⬚ **45m Food Science Lessons - Lesson 28** *The Science of Baking* Materials: Elements of Baking **→ ANALYZE** • Review how the bake went, reread, and plan any adjustments that you want to try next week OR • Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.	
WEEK 6	⬚ **45m Food Science Lessons - Lesson 29** *The Science of Baking* Materials: Elements of Baking **→ READ, NARRATE, & DISCUSS** Ch.8 p.393-398	
WEEK 6	⬚ **45m Food Science Lessons - Lesson 30** *The Science of Baking* Materials: Elements of Baking **→ REREAD & DISCUSS** Ch.3. Check that the ideas make sense now that you have some experience. **→ PLAN** Decide on your Cooking Lesson recipe for next week, or wait to decide until after dinner night. Add to shopping list, as needed.	
WEEK 6	⬚ **60m Food Science Labs - Lesson 6** Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies **→ KITCHEN LAB** Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 7	⬚ **45m Food Science Lessons - Lesson 31** *The Science of Baking* Materials: Elements of Baking **→ READ, NARRATE, & DISCUSS** Ch.9 p.427-433	• **PLAN WEEKLY** ⬚ take a nature walk ⬚ science free read
WEEK 7	⬚ **45m Food Science Lessons - Lesson 32** *The Science of Baking* Materials: Elements of Baking, indicated ingredients, and kitchen supplies **→ BAKING DAY!** Prepare your selected recipe. Record what you do and what you observe in your notebook.	
WEEK 7	⬚ **45m Food Science Lessons - Lesson 33** *The Science of Baking* Materials: Elements of Baking **→ ANALYZE** • Review how the bake went, reread, and plan any adjustments that you want to try next week OR • Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.	
WEEK 7	⬚ **45m Food Science Lessons - Lesson 34** *The Science of Cooking* Materials: Salt, Fat, Acid, Heat **→ READ, NARRATE, & DISCUSS** p.101-110 ∞ Video Link: America's Test Kitchen ∞ Video Link: Alveary Chemistry Companion	

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Term 1

WEEK 7

⬚ **45m Food Science Lessons - Lesson 35**
The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS
p.111-119

→ PLAN
Decide on your Cooking Lesson recipe for next week, or wait to decide until after dinner night. Add to shopping list, as needed.

WEEK 7

⬚ **60m Food Science Labs - Lesson 7**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 8

⬚ **45m Food Science Lessons - Lesson 36**
The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS
p.120-129

- **PLAN WEEKLY**

⬚ take a nature walk
⬚ science free read

WEEK 8

⬚ **45m Food Science Lessons - Lesson 37**
The Science of Baking

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 8

⬚ **45m Food Science Lessons - Lesson 38**
The Science of Baking

Materials: Elements of Baking

→ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.

WEEK 8

⬚ **45m Food Science Lessons - Lesson 39**
The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS
p.238-239

→ ANALYZE
Spend some time comparing different recipes in SFAH to see how Salt, Fat, and Acid work together before we examine Heat. Rather than another material ingredient, Heat stands apart because what we are adding is energy.

WEEK 8

⬚ **45m Food Science Lessons - Lesson 40**
The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ ANALYZE
Compare different recipes in SFAH to see how Salt, Fat, and Acid work together.

→ PLAN
Decide on your Cooking Lesson recipe for next week, or wait to decide until after dinner night. Add to shopping list, as needed.

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Term 1

WEEK 8

⬚ **60m Food Science Labs - Lesson 8**

▯ Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 9

⬚ **45m Food Science Lessons - Lesson 41**
The Science of Cooking

▯ Materials: Salt, Fat, Acid, Heat

➔ READ, NARRATE, & DISCUSS
p.131-139

∞ Video Link: Alveary Chemistry Companion

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 9

⬚ **45m Food Science Lessons - Lesson 42**
The Science of Baking

▯ Materials: Elements of Baking, indicated ingredients, and kitchen supplies

➔ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 9

⬚ **45m Food Science Lessons - Lesson 43**
The Science of Baking

▯ Materials: Elements of Baking

➔ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.

WEEK 9

⬚ **45m Food Science Lessons - Lesson 44**
The Science of Cooking

▯ Materials: Salt, Fat, Acid, Heat

➔ READ, NARRATE, & DISCUSS
p.140-150

WEEK 9

⬚ **45m Food Science Lessons - Lesson 45**
The Science of Cooking

▯ Materials: Salt, Fat, Acid, Heat

➔ READ, NARRATE, & DISCUSS
p.151-160

➔ PLAN
Decide on your Cooking Lesson recipe for next week, or wait to decide until after dinner night. Add to shopping list, as needed.

WEEK 9

⬚ **60m Food Science Labs - Lesson 9**

▯ Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 10

⬚ **45m Food Science Lessons - Lesson 46**
The Science of Cooking

▯ Materials: Salt, Fat, Acid, Heat

➔ READ, NARRATE, & DISCUSS
p.161-170 (end of Reducing)

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

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WEEK 10

⬚ **45m Food Science Lessons - Lesson 47**
The Science of Baking

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

➔ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 10

⬚ **45m Food Science Lessons - Lesson 48**
The Science of Baking

Materials: Elements of Baking

➔ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.

WEEK 10

⬚ **45m Food Science Lessons - Lesson 49**
The Science of Cooking

Materials: Salt, Fat, Acid, Heat

➔ READ, NARRATE, & DISCUSS
p.170-180 (end of Baking)

WEEK 10

⬚ **45m Food Science Lessons - Lesson 50**
The Science of Cooking

Materials: Salt, Fat, Acid, Heat

➔ READ, NARRATE, & DISCUSS
p.180-190

➔ PLAN
Decide on your Cooking Lesson recipe for next week, or wait to decide until after dinner night. Add to shopping list, as needed.

WEEK 10

⬚ **60m Food Science Labs - Lesson 10**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 11

⬚ **45m Food Science Lessons - Lesson 51**
The Science of Cooking

Materials: Salt, Fat, Acid, Heat

➔ READ, NARRATE, & DISCUSS
p.191-200

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 11

⬚ **45m Food Science Lessons - Lesson 52**
The Science of Baking

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

➔ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 11

⬚ **45m Food Science Lessons - Lesson 53**
The Science of Baking

Materials: Elements of Baking

➔ ANALYZE
• Review how the bake went and consider what you learned.

WEEK 11

⬚ **45m Food Science Lessons - Lesson 54**
Present What You've Learned

➔ PREPARE
For one of your exam questions, you will be asked to give an informal presentation to your

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teacher on something you have learned about during lessons this term, such as how a particular recipe or ingredient works. (This is distinct from what you learned by preparing meals on dinner night.) Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like. If you want to create something to present, use this time to work on it.

WEEK 11

⬚ **45m Food Science Lessons - Lesson 55**
Present What You've Learned

➔ PREPARE
For one of your exam questions, you will be asked to give an informal presentation to your teacher on something you have learned about during lessons this term, such as how a particular recipe or ingredient works. (This is distinct from what you learned by preparing meals on dinner night.) Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like. If you want to create something to present, use this time to work on it.

WEEK 11

⬚ **60m Food Science Labs - Lesson 11**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 56**
Exam Week

➔ EXAMS
Answer question(s) related to the course.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 57**
Exam Week

➔ EXAMS
Answer question(s) related to the course.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 58**
Exam Week

➔ EXAMS
Answer question(s) related to the course.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 59**
Exam Week

➔ EXAMS
Answer question(s) related to the course.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 60**
Exam Week

➔ EXAMS
Answer question(s) related to the course.

WEEK 12

⬚ **60m Food Science Labs - Lesson 12**
Exam: Dinner Night

➔ EXAMS
Answer exam questions from:
Kitchen Lab experience

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Term 2

WEEK 1

⬚ **45m Food Science Lessons - Lesson 61**
Preparation

▯ Materials: internet access, favorite recipes & cookbooks

→ INTRO
Now it's time to make what you learned in Term 1 entirely your own and apply it more broadly. Terms 2 & 3 will be entirely application. Schedule your baking day and dinner night, but otherwise, spend time comparing recipes, watching any videos (below and in your Quick Links) that you find interesting and helpful, and thinking about how the ideas apply - don't forget to get your shopping list to your teacher on time!

∞ Video Link: America's Test Kitchen has an extensive video library on YouTube (the playlist included in term 1 a few times is called What's Eating Dan)
∞ Video Link: The Loopy Whisk is by the baker and chemist who wrote The Elements of Baking
∞ Video Link: Samin Nosrat Cooking features the chef who wrote Salt Fat Acid Heat

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 1

⬚ **45m Food Science Lessons - Lesson 62**
The Science of Baking

▯ Materials: Elements of Baking

→ PLAN
Consider what baking lessons you would like to learn this term; what recipes you would like to experiment with (in this book or from any other source); what ingredients you would like to better understand. Set a goal for this term and plan using the skills you have learned.

WEEK 1

⬚ **45m Food Science Lessons - Lesson 63**
The Science of Cooking

▯ Materials: Salt, Fat, Acid, Heat

→ PLAN
Consider what cooking lessons you would like to learn this term; what recipes you would like to experiment with (in this book or from any other source); what ingredients you would like to better understand. Set a goal for this term and plan using the skills you have learned.

WEEK 1

⬚ **45m Food Science Lessons - Lesson 64**
Literature of Food Science & Culture

▯ Materials: culinary anthropology selection

→ INTRO
This term, you will explore how food influences people and how people influence food. You might choose a culture you already know, but would like to better appreciate. You might choose a new culture that you'd like to meet. You might even dabble in more than one culture. The video linked is provided as inspiration:
∞ Video Link: How Pancakes are Made in 11 Countries

→ READ, NARRATE, & DISCUSS
Read at your own pace, as each selection and your interest in them will vary. You could choose to read more during your flex day, during your weekly free read time, or on your own at the end of term. If you choose not to finish the book, that is fine, too. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

WEEK 1

⬚ **45m Food Science Lessons - Lesson 65**
Flex Day

▯ Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

→ NOTE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

Read more from your culinary anthropology selection.

WEEK 1

⬚ **60m Food Science Labs - Lesson 13**

▯ Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping

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Term 2

list, if you haven't already.

WEEK 2

⬚ **45m Food Science Lessons - Lesson 66**
Preparation

▯ Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

➔ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY
⬚ take a nature walk
⬚ science free read

WEEK 2

⬚ **45m Food Science Lessons - Lesson 67**
The Science of Baking

▯ Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

➔ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 2

⬚ **45m Food Science Lessons - Lesson 68**
The Science of Baking

▯ Materials: Elements of Baking, selected recipe

➔ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 2

⬚ **45m Food Science Lessons - Lesson 69**
Literature of Food Science & Culture

▯ Materials: culinary anthropology selection

➔ READ, NARRATE, & DISCUSS
Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

WEEK 2

⬚ **45m Food Science Lessons - Lesson 70**
Flex Day

▯ Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

➔ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

➔ READ, NARRATE, & DISCUSS
Current Events from the links (below and in your Quick Links):
∞ Website Link: Science News Explores
∞ Website Link: Food & Wine

OR

Read more from your culinary anthropology selection.

WEEK 2

⬚ **60m Food Science Labs - Lesson 14**

▯ Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

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WEEK 3

⬚ **45m Food Science Lessons - Lesson 71**

Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY
⬚ take a nature walk
⬚ science free read

WEEK 3

⬚ **45m Food Science Lessons - Lesson 72**

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 3

⬚ **45m Food Science Lessons - Lesson 73**

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 3

⬚ **45m Food Science Lessons - Lesson 74**

Literature of Food Science & Culture

Materials: culinary anthropology selection

→ READ, NARRATE, & DISCUSS
Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

WEEK 3

⬚ **45m Food Science Lessons - Lesson 75**

Flex Day

Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

→ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don’t forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read more from your culinary anthropology selection.

WEEK 3

⬚ **60m Food Science Labs - Lesson 15**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven’t already.

WEEK 4

⬚ **45m Food Science Lessons - Lesson 76**

Preparation

Materials: Elements of Baking, Salt Fat Acid Heat, selected recipes

→ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY
⬚ take a nature walk
⬚ science free read

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WEEK 4

⬚ **45m Food Science Lessons - Lesson 77**

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 4

⬚ **45m Food Science Lessons - Lesson 78**

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 4

⬚ **45m Food Science Lessons - Lesson 79**

Literature of Food Science & Culture

Materials: culinary anthropology selection

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS
Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

★ TEACHER TIP
Students are scheduled to complete a citizen science project next week during which they will audit food waste. The project preparation begins in the next lesson. This project can be moved to any other week during the term based on interest or convenience.

WEEK 4

⬚ **45m Food Science Lessons - Lesson 80**

Materials: internet access

→ INTRO
Let's think about how we interact with food as citizens. Begin with a citizen science project about food waste. You will report on this project toward the end of the term.

→ PREPARE
Follow the link and explore the website to learn about the project. Scroll down to "Follow these steps to get started." Use today to prepare for the project. You will take a break from baking and cooking to complete the project next week. If you prefer, you may move this project to any other week this term.
∞ Website Link: Food Conservation

WEEK 4

⬚ **60m Food Science Labs - Lesson 16**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

Prep: Read Teacher Tip

→ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

★ TEACHER TIP
Students are conducting citizen science next week and will not have their regular dinner night. They may need to adjust their schedule, so that their Kitchen Waste Audit is completed at the end of the week.

WEEK 5

⬚ **45m Food Science Lessons - Lesson 81**

Materials: internet access, kitchen scale

→ DATA COLLECTION
Complete the first Kitchen Waste Audit
∞ Website Link: Food Conservation

• PLAN WEEKLY
⬚ take a nature walk
⬚ science free read

WEEK 5

⬚ **45m Food Science Lessons - Lesson 82**

Materials: internet access, kitchen scale

→ DATA COLLECTION
Conduct Day 1 Food Waste Log
∞ Website Link: Food Conservation

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WEEK 5

⬚ **45m Food Science Lessons - Lesson 83**

Materials: internet access, kitchen scale

→ DATA COLLECTION
Conduct Day 2 Food Waste Log
∞ Website Link: Food Conservation

WEEK 5

⬚ **45m Food Science Lessons - Lesson 84**

Materials: internet access, kitchen scale

→ DATA COLLECTION
Conduct Day 3 Food Waste Log
∞ Website Link: Food Conservation

WEEK 5

⬚ **45m Food Science Lessons - Lesson 85**

Materials: internet access, kitchen scale

→ DATA COLLECTION
Conduct Day 4 Food Waste Log
∞ Website Link: Food Conservation

WEEK 5

⬚ **60m Food Science Labs - Lesson 17**

Materials: internet access, kitchen scale

→ DATA COLLECTION
Conduct the second Kitchen Waste Audit. Adjust your schedule, if needed, so that this audit is completed after all 4 Daily Food Waste Logs.
∞ Website Link: Food Conservation

WEEK 6

⬚ **45m Food Science Lessons - Lesson 86**
Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 6

⬚ **45m Food Science Lessons - Lesson 87**
The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 6

⬚ **45m Food Science Lessons - Lesson 88**
The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 6

⬚ **45m Food Science Lessons - Lesson 89**
Literature of Food Science & Culture

Materials: culinary anthropology selection

→ READ, NARRATE, & DISCUSS
Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

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WEEK 6

☐ **45m Food Science Lessons - Lesson 90**
Flex Day

Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

→ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read more from your culinary anthropology selection.

WEEK 6

☐ **60m Food Science Labs - Lesson 18**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 7

☐ **45m Food Science Lessons - Lesson 91**
Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• **PLAN WEEKLY**
☐ take a nature walk
☐ science free read

WEEK 7

☐ **45m Food Science Lessons - Lesson 92**
The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 7

☐ **45m Food Science Lessons - Lesson 93**
The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 7

☐ **45m Food Science Lessons - Lesson 94**
Literature of Food Science & Culture

Materials: culinary anthropology selection

→ READ, NARRATE, & DISCUSS
Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

WEEK 7

☐ **45m Food Science Lessons - Lesson 95**
Flex Day

Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

→ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

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OR

→ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read more from your culinary anthropology selection.

WEEK 7

⬚ **60m Food Science Labs - Lesson 19**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 8

⬚ **45m Food Science Lessons - Lesson 96**
Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY
⬚ take a nature walk
⬚ science free read

WEEK 8

⬚ **45m Food Science Lessons - Lesson 97**
The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 8

⬚ **45m Food Science Lessons - Lesson 98**
The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 8

⬚ **45m Food Science Lessons - Lesson 99**
Literature of Food Science & Culture

Materials: culinary anthropology selection

→ READ, NARRATE, & DISCUSS
Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

WEEK 8

⬚ **45m Food Science Lessons - Lesson 100**
Flex Day

Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

→ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read more from your culinary anthropology selection.

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WEEK 8

⬚ **60m Food Science Labs - Lesson 20**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 9

⬚ **45m Food Science Lessons - Lesson 101**
Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

➔ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 9

⬚ **45m Food Science Lessons - Lesson 102**
The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

➔ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 9

⬚ **45m Food Science Lessons - Lesson 103**
The Science of Baking

Materials: Elements of Baking, selected recipe

➔ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 9

⬚ **45m Food Science Lessons - Lesson 104**
Literature of Food Science & Culture

Materials: culinary anthropology selection

➔ READ, NARRATE, & DISCUSS
Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

WEEK 9

⬚ **45m Food Science Lessons - Lesson 105**
Flex Day

Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

➔ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

➔ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read more from your culinary anthropology selection.

WEEK 9

⬚ **60m Food Science Labs - Lesson 21**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

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Term 2

WEEK 10	⬚ **45m Food Science Lessons - Lesson 106** *Preparation* ▯ Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes **➔ PREPARE** Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.	**• PLAN WEEKLY** ⬚ take a nature walk ⬚ science free read
WEEK 10	⬚ **45m Food Science Lessons - Lesson 107** *The Science of Baking* ▯ Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies **➔ BAKING DAY!** Prepare your selected recipe. Record what you do and what you observe in your notebook.	
WEEK 10	⬚ **45m Food Science Lessons - Lesson 108** *The Science of Baking* ▯ Materials: Elements of Baking, selected recipe **➔ ANALYZE** • Review how the bake went, reread, and plan any adjustments that you want to try next week OR • Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.	
WEEK 10	⬚ **45m Food Science Lessons - Lesson 109** *Literature of Food Science & Culture* ▯ Materials: culinary anthropology selection **➔ READ, NARRATE, & DISCUSS** Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.	
WEEK 10	⬚ **45m Food Science Lessons - Lesson 110** *Flex Day* ▯ Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection **➔ EXPLORE** Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time! OR **➔ READ, NARRATE, & DISCUSS** Current Events from Science News Explores or Food & Wine (in your Quick Links) OR Read more from your culinary anthropology selection.	
WEEK 10	⬚ **60m Food Science Labs - Lesson 22** ▯ Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies **➔ KITCHEN LAB** Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 11	⬚ **45m Food Science Lessons - Lesson 111** *Preparation* ▯ Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes **➔ PREPARE** Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.	**• PLAN WEEKLY** ⬚ take a nature walk ⬚ science free read

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Term 2

WEEK 11

⬚ **45m Food Science Lessons - Lesson 112**
The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 11

⬚ **45m Food Science Lessons - Lesson 113**
The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE
• Review how the bake went and consider what you learned.

WEEK 11

⬚ **45m Food Science Lessons - Lesson 114**
Present What You've Learned

→ PREPARE
For one of your exam questions, you will be asked to give an informal presentation to your teacher on your citizen science project. This can be an oral presentation (e.g., a poster presentation, PowerPoint slides, etc) or something you've prepared (e.g., an essay, Canva brochure, or graphic, painting, etc). Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like.

WEEK 11

⬚ **45m Food Science Lessons - Lesson 115**
Present What You've Learned

→ PREPARE
For one of your exam questions, you will be asked to give an informal presentation to your teacher on your citizen science project. This can be an oral presentation (e.g., a poster presentation, PowerPoint slides, etc) or something you've prepared (e.g., an essay, Canva brochure, or graphic, painting, etc). Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like.

WEEK 11

⬚ **60m Food Science Labs - Lesson 23**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven’t already.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 116**
Exam Week

→ EXAMS
Answer question(s) related to the course.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 117**
Exam Week

→ EXAMS
Answer question(s) related to the course.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 118**
Exam Week

→ EXAMS
Answer question(s) related to the course.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 119**
Exam Week

→ EXAMS
Answer question(s) related to the course.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 120**
Exam Week

→ EXAMS
Answer question(s) related to the course.

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WEEK 12	⬚ **60m Food Science Labs - Lesson 24** *Exam: Dinner Night* **→ EXAMS** Answer exam questions from: Kitchen Lab experience

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Term 3

WEEK 1

⬚ **45m Food Science Lessons - Lesson 121**
Preparation

▯ Materials: internet access, favorite recipes & cookbooks

➔ INTRO
Schedule your baking day and dinner night, but otherwise, spend time comparing recipes, watching any videos (below and in your Quick Links) that you find interesting and helpful, and thinking about how the ideas apply - don’t forget to get your shopping list to your teacher on time!

∞ Video Link: America's Test Kitchen has an extensive video library on YouTube (the playlist included in term 1 a few times is called What’s Eating Dan)
∞ Video Link: The Loopy Whisk is by the baker and chemist who wrote The Elements of Baking
∞ Video Link: Samin Nosrat Cooking features the chef who wrote Salt, Fat, Acid, Heat

• PLAN WEEKLY
⬚ take a nature walk
⬚ science free read

WEEK 1

⬚ **45m Food Science Lessons - Lesson 122**
The Science of Baking

▯ Materials: Elements of Baking

➔ PLAN
Consider what lessons you would like to learn this term; what recipes you would like to experiment with (in this book or from any other source); what ingredients you would like to better understand. Set a goal for this term and plan using the skills you have learned.

WEEK 1

⬚ **45m Food Science Lessons - Lesson 123**
The Science of Cooking

▯ Materials: Salt, Fat, Acid, Heat

➔ PLAN
Consider what cooking lessons you would like to learn this term; what recipes you would like to experiment with (in this book or from any other source); what ingredients you would like to better understand. Set a goal for this term and plan using the skills you have learned.

WEEK 1

⬚ **45m Food Science Lessons - Lesson 124**
Literature of Food Science & Culture

▯ Materials: Culinary Reactions

➔ READ, NARRATE, & DISCUSS
About 1 chapter. You won't have time to finish this book in this one lesson slot for the term. You could choose to read more during your flex day, during your weekly free read time, or on your own at the end of term. If you choose not to finish the book, that is fine, too.

WEEK 1

⬚ **45m Food Science Lessons - Lesson 125**
Flex Day

▯ Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

➔ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don’t forget to get your shopping list to your teacher on time!

OR

➔ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 1

⬚ **60m Food Science Labs - Lesson 25**

▯ Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven’t already.

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Term 3

WEEK 2

⬚ **45m Food Science Lessons - Lesson 126**
Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 2

⬚ **45m Food Science Lessons - Lesson 127**
The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 2

⬚ **45m Food Science Lessons - Lesson 128**
The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 2

⬚ **45m Food Science Lessons - Lesson 129**
Literature of Food Science & Culture

Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS
About 1 chapter

WEEK 2

⬚ **45m Food Science Lessons - Lesson 130**
Flex Day

Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don’t forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 2

⬚ **60m Food Science Labs - Lesson 26**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven’t already.

WEEK 3

⬚ **45m Food Science Lessons - Lesson 131**
Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

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Term 3

WEEK 3

⬚ **45m Food Science Lessons - Lesson 132**
The Science of Baking

▯ Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 3

⬚ **45m Food Science Lessons - Lesson 133**
The Science of Baking

▯ Materials: Elements of Baking, selected recipe

→ ANALYZE
• Review how the bake went, reread, plan any adjustments that you want to try in 2 weeks (there is a project scheduled next week) OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 3

⬚ **45m Food Science Lessons - Lesson 134**
Literature of Food Science & Culture

▯ Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS
About 1 chapter

WEEK 3

⬚ **45m Food Science Lessons - Lesson 135**
Flex Day

▯ Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 3

⬚ **60m Food Science Labs - Lesson 27**

▯ Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 4

⬚ **45m Food Science Lessons - Lesson 136**
Preparation

▯ Materials: Salt, Fat, Acid, Heat, selected recipe

→ PREPARE
Read over the week's dinner recipe. Consider how the recipe works and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook, gather ingredients and supplies, or get started on your research project.

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 4

⬚ **45m Food Science Lessons - Lesson 137**

▯ Materials: internet or library access

→ INTRO
Last term, we evaluated food waste in our community. This is one important aspect of how we interact with food as citizens. Another is being aware of where our food comes from and how the production and process of getting that food to our plates impacts the world around us. Watch the linked video to begin thinking about the problem. While there is no one-size-fits-all approach, we want to make decisions that do the most good and the least harm. You will have three periods for this research and will prepare a composition about your findings.
∞ Video Link: What is the true price of the food we eat?

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Term 3

→ RESEARCH
Select 3 ingredients from your last dinner recipe. For each ingredient, try to answer these questions:
• How is this ingredient produced? Is it grown on farms? Is it further processed in a factory?
• Where is it produced?
• How is it packaged and distributed?
• What kind of waste is generated from production, packaging, and distribution?
• How does the whole process positively and negatively affect people and the environment?
• What systems in society support the use of this energy source (e.g., economic, transportation, etc.)?

WEEK 4

☐ 45m Food Science Lessons - Lesson 138

Materials: internet or library access

→ RESEARCH
Continue researching food production.

WEEK 4

☐ 45m Food Science Lessons - Lesson 139

Materials: internet or library access

→ RESEARCH
Complete your research. Use any remaining time to begin thinking about your composition. Effective communication of complex ideas is a major challenge in science, so think about how you communicate best and plan your composition thoughtfully! You could compose an expository essay, a PowerPoint or poster presentation, a graphic display using Canva or watercolor, or any other form of expression that allows you to communicate what you have learned effectively.

WEEK 4

☐ 45m Food Science Lessons - Lesson 140
Literature of Food Science & Culture

Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS
About 1 chapter

WEEK 4

☐ 60m Food Science Labs - Lesson 28

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven’t already.

WEEK 5

☐ 45m Food Science Lessons - Lesson 141
Preparation

Materials: Elements of Baking, Salt Fat Acid Heat, selected recipes

→ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY
☐ take a nature walk
☐ science free read

WEEK 5

☐ 45m Food Science Lessons - Lesson 142
The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 5

☐ 45m Food Science Lessons - Lesson 143
The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

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WEEK 5

⬚ **45m Food Science Lessons - Lesson 144**
Literature of Food Science & Culture

Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS
About 1 chapter

WEEK 5

⬚ **45m Food Science Lessons - Lesson 145**
Flex Day

Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don’t forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 5

⬚ **60m Food Science Labs - Lesson 29**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven’t already.

WEEK 6

⬚ **45m Food Science Lessons - Lesson 146**
Preparation

Materials: Elements of Baking, Salt Fat Acid Heat, selected recipes

→ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY
⬚ take a nature walk
⬚ science free read

WEEK 6

⬚ **45m Food Science Lessons - Lesson 147**
The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 6

⬚ **45m Food Science Lessons - Lesson 148**
The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 6

⬚ **45m Food Science Lessons - Lesson 149**
Literature of Food Science & Culture

Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS
About 1 chapter

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Term 3

WEEK 6

⬚ **45m Food Science Lessons - Lesson 150**

Flex Day

Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 6

⬚ **60m Food Science Labs - Lesson 30**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 7

⬚ **45m Food Science Lessons - Lesson 151**

Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 7

⬚ **45m Food Science Lessons - Lesson 152**

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 7

⬚ **45m Food Science Lessons - Lesson 153**

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 7

⬚ **45m Food Science Lessons - Lesson 154**

Literature of Food Science & Culture

Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS
About 1 chapter

WEEK 7

⬚ **45m Food Science Lessons - Lesson 155**

Flex Day

Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

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Term 3

➔ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 7

⬚ 60m Food Science Labs - Lesson 31

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven’t already.

WEEK 8

⬚ 45m Food Science Lessons - Lesson 156
Preparation

Materials: Elements of Baking, Salt Fat Acid Heat, selected recipes

➔ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 8

⬚ 45m Food Science Lessons - Lesson 157
The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

➔ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 8

⬚ 45m Food Science Lessons - Lesson 158
The Science of Baking

Materials: Elements of Baking, selected recipe

➔ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 8

⬚ 45m Food Science Lessons - Lesson 159
Literature of Food Science & Culture

Materials: Culinary Reactions

➔ READ, NARRATE, & DISCUSS
About 1 chapter

WEEK 8

⬚ 45m Food Science Lessons - Lesson 160
Flex Day

Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

➔ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don’t forget to get your shopping list to your teacher on time!

OR

➔ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

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Term 3

WEEK 8

⬚ **60m Food Science Labs - Lesson 32**

▯ Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 9

⬚ **45m Food Science Lessons - Lesson 161**
Preparation

▯ Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

➔ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 9

⬚ **45m Food Science Lessons - Lesson 162**
The Science of Baking

▯ Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

➔ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 9

⬚ **45m Food Science Lessons - Lesson 163**
The Science of Baking

▯ Materials: Elements of Baking, selected recipe

➔ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 9

⬚ **45m Food Science Lessons - Lesson 164**
Literature of Food Science & Culture

▯ Materials: Culinary Reactions

➔ READ, NARRATE, & DISCUSS
About 1 chapter

WEEK 9

⬚ **45m Food Science Lessons - Lesson 165**
Flex Day

▯ Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

➔ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

➔ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 9

⬚ **60m Food Science Labs - Lesson 33**

▯ Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

Science: Food Science

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Term 3

WEEK 10

⬚ **45m Food Science Lessons - Lesson 166**
Preparation

Materials: Elements of Baking, Salt Fat Acid Heat, selected recipes

➔ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 10

⬚ **45m Food Science Lessons - Lesson 167**
The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

➔ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 10

⬚ **45m Food Science Lessons - Lesson 168**
The Science of Baking

Materials: Elements of Baking, selected recipe

➔ ANALYZE
• Review how the bake went, reread, and plan any adjustments that you want to try next week
OR
• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 10

⬚ **45m Food Science Lessons - Lesson 169**
Literature of Food Science & Culture

Materials: Culinary Reactions

➔ READ, NARRATE, & DISCUSS
About 1 chapter

WEEK 10

⬚ **45m Food Science Lessons - Lesson 170**
Flex Day

Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

➔ EXPLORE
Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

➔ READ, NARRATE, & DISCUSS
Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 10

⬚ **60m Food Science Labs - Lesson 34**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

➔ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 11

⬚ **45m Food Science Lessons - Lesson 171**
Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

➔ PREPARE
Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

Science: Food Science

Click THIS text or scan the QR code for links.

QR

Term 3

WEEK 11

⬚ **45m Food Science Lessons - Lesson 172**
The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!
Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 11

⬚ **45m Food Science Lessons - Lesson 173**
The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE
- Review how the bake went and consider what you learned.

WEEK 11

⬚ **45m Food Science Lessons - Lesson 174**
Present What You've Learned

→ PREPARE
For one of your exam questions, you will be asked to give an informal presentation to your teacher on your citizen science research. This can be an oral presentation (e.g., a poster presentation, PowerPoint slides, etc.) or something you've prepared (e.g., an essay, Canva brochure, or graphic, painting, etc.). Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like.

WEEK 11

⬚ **45m Food Science Lessons - Lesson 175**
⬚ *45m Food Science - Lesson 175*
Present What You've Learned

→ PREPARE
For one of your exam questions, you will be asked to give an informal presentation to your teacher on your citizen science research. This can be an oral presentation (e.g. a poster presentation, Power Point slides, etc) or something you've prepared (e.g. an essay, Canva brochure or graphic, painting, etc). Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like.

WEEK 11

⬚ **60m Food Science Labs - Lesson 35**

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB
Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven’t already.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 176**
Exam Week

→ EXAMS
Answer question(s) related to the course.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 177**
Exam Week

→ EXAMS
Answer question(s) related to the course.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 178**
Exam Week

→ EXAMS
Answer question(s) related to the course.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 179**
Exam Week

→ EXAMS
Answer question(s) related to the course.

WEEK 12

⬚ **45m Food Science Lessons - Lesson 180**
Exam Week

→ EXAMS
Answer question(s) related to the course.

Science: Food Science

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Term 3

WEEK 12

⬚ **60m Food Science Labs - Lesson 36**
Exam: Dinner Night

→ EXAMS
Answer exam questions from:
Kitchen Lab experience

Science: Food Science

Examination

Term 1

Food Science Lessons

· Salt Fat Acid Heat: Tell something you learned about the chemistry of salt OR fat (lipids) OR acid, and the observations you made about this ingredient in the kitchen.

· Salt, Fat, Acid, Heat: Tell something you learned about what heat is and what it does physically/chemically. Also, tell about the observations you made about heat in the kitchen.

· Elements of Baking: Tell about one 'element of baking' that you read about this term, including what properties it contributes to a bake.

· Tell about a meal you prepared for 'dinner night' and explain how it was made. You may refer to the recipe to help, but you should be able to explain HOW to prepare the food and tell about how salt/acid/fat/heat contribute to the success of the dish. Share your lab notebook with your teacher.

· Give an informal presentation to your teacher on something you have learned about this term, such as how a particular recipe or ingredient works.

Term 2

Food Science Lessons

· Share your lab notebook with your teacher and explain one of your bakes. What worked/didn't? How did ingredient substitutions affect the product?

· Culinary Anthropology selection: Tell what you learned about food in your selected culture. How are salt, fat, acid, and heat used by your selected culture?

· Tell something that stood out to you about a chosen recipe, current event, OR your culinary anthropology selection.

· Tell about a meal you prepared for 'dinner night' and explain how it was made. You may refer to the recipe to help, but you should be able to explain HOW to prepare the food and tell about how salt/acid/fat/heat contribute to the success of the dish. Share your notebook with your teacher.

· Give an informal presentation to your teacher on your citizen science project.

Term 3

Food Science Lessons

· Share your lab notebook with your teacher and explain one of your bakes. What worked/didn't? How did ingredient substitutions affect the product? How does notebooking impact your learning experience?

· Explain an idea that you learned about in Culinary Reactions.

· Tell something that stood out to you about a chosen recipe, current event, OR Culinary Reactions.

· Tell about a meal you prepared for 'dinner night' and explain how it was made. You may refer to the recipe to help, but you should be able to explain HOW to prepare the food and tell about how salt/acid/fat/heat contribute to the success of the dish. Share your notebook with your teacher.

· Give an informal presentation to your teacher on your citizen science research.